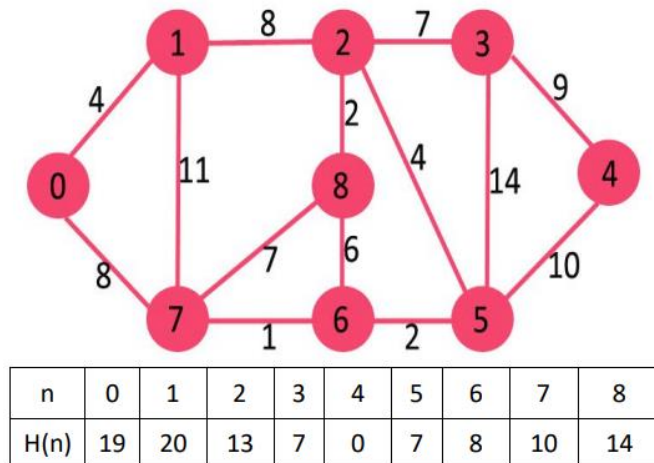


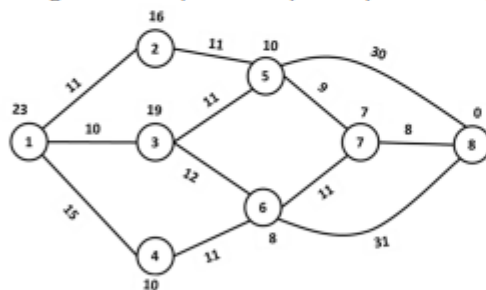
1. Explain in detail the architecture of a goal based agent and learning agent.
2. With suitable examples, explain the working of the following algorithms.
 - a) Gradient descent [4]
 - b) Local Beam Search [3]
 - c) Stochastic Beam search [3]

Perform A* search algorithm on the graph below to find the shortest path from node 0 to node 4. Each edge is labeled by the cost to traverse that edge. Heuristic values are given in the table below the graph.



1. a) Develop a PEAS description of the following task environments: [5]
 - i. Robot soccer player
 - ii. Internet book shopping agent
 - iii. Medical diagnosis system
 - iv. Refinery controller
 - v. Autonomous Mars rover
- b) Discuss the importance of AI in defense applications. [5]
2. Explain the methodology, advantages and disadvantages of the following search algorithms with suitable illustrations. [10]
 - Hill climbing
 - Genetic algorithm
 - Simulated annealing

3. Compare and contrast various agent programs based on the following modalities: Architecture, Functionality, Complexity, Task execution, Applications. [10]
4. Consider the following network in which the source node is 1 and the destination node is 8. The nodes represent different cities and the edge weights represent the time taken to travel between a pair of cities. The weights mentioned above the nodes represent the estimated time taken by the nodes to reach the destination directly. Determine the optimal time to reach the destination from source using A* algorithm. Also prove the optimality of the A* algorithm. [10]



5. The iterative lengthening search is defined as an iterative analog of uniform cost search. The idea is to use increasing limits on path cost. If a node is generated whose path cost exceeds the current limit, it is immediately discarded. For each new iteration, the limit is set to the lowest path cost of any node discarded in the previous iteration. [10]

Develop the pseudocode for iterative lengthening search. Apply it to the following instance of the 8-puzzle problem and compare its performance with uniform cost search.

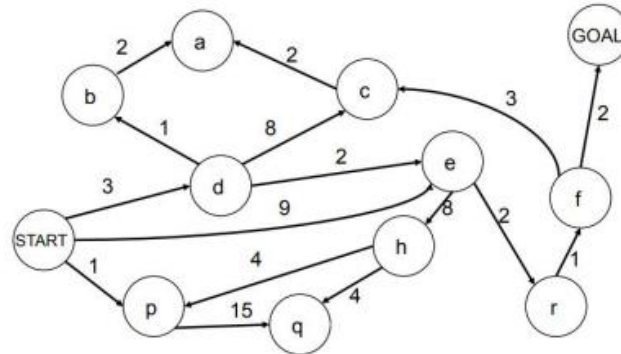
7	2	4
5		6
8	3	1

Start State

	1	2
3	4	5
6	7	8

Goal State

- For the following activities, provide the description of the task environments.
 - Playing a tennis match
 - Chess with a clock
- Describe model based agent and learning based agent with necessary diagrams.
- Illustrate DFS and uniform cost search for the following graph. Compare the search strategies based on the evaluation criteria.



- Perform A* search and Greedy best first search on the given graph to find the shortest path from node S to node G. Explicitly write down the queue at each step. Compare the two on the basis of optimality and completeness.

